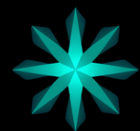


Loan Approval Prediction Using Machine Learning



Predicting Loan Approval Using Data Analytics & Machine Learning



PYTHON | PANDAS | NUMPY | SCIKIT-LEARN

MATPLOTLIB | SEABORN | ML ANALYTICS

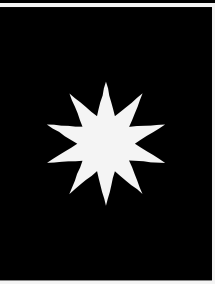
Presented by Atul Soni

Business Problem Statement

The Challenge

Banks and financial institutions process thousands of loan applications every day, making manual evaluation time-consuming, costly, and prone to human bias. The challenge is to accurately identify creditworthy applicants while minimizing the risk of loan defaults. This project leverages Machine Learning to automate the loan approval process by analyzing applicant financial, credit, and asset-related information, enabling faster, more consistent, and data-driven lending decisions.





Dataset Overview

Features & Target Variable

Features (Independent Variables)

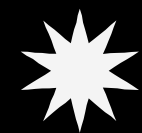
The model uses the following applicant attributes to predict loan approval:

- Number of Dependents
- Education Level
- Self-Employed Status
- Annual Income
- Loan Amount

- Loan Term
- CIBIL Score
- Residential Assets Value
- Commercial Assets Value
- Luxury Assets Value
- Bank Asset Value

Target Variable (Dependent Variable)

- Loan Status
- Approved (1)
- Rejected (0)





ML Pipeline Workflow



1

Data Collection

Gather & clean raw data

2

Exploratory Analysis

Discover patterns & insights

3

Feature Engineering

Create & preprocess features

4

Train-Test Split

80:20 split & StandardScaler

5

Model & Evaluate

Logistic Regression metrics

Data Cleaning & Preprocessing



Data Preparation Steps

Dropped non-predictive Loan ID column, cleaned and standardized all column names, checked for and handled missing values, and prepared a structured dataset ready for analysis.

Remove Loan ID

Standardize Names

Handle Missing Values

Structure Dataset



Exploratory Data Analysis

Key Data Insights & Patterns

Loan Amount Distribution:

Nearly 75% of loan applications have a loan amount below ₹2 crore, indicating demand is concentrated in the lower-to-mid loan range.

CIBIL Score Distribution:

Applicant CIBIL scores span from 300 to 900, with the majority clustered between 500 and 800, reflecting diverse credit profiles.

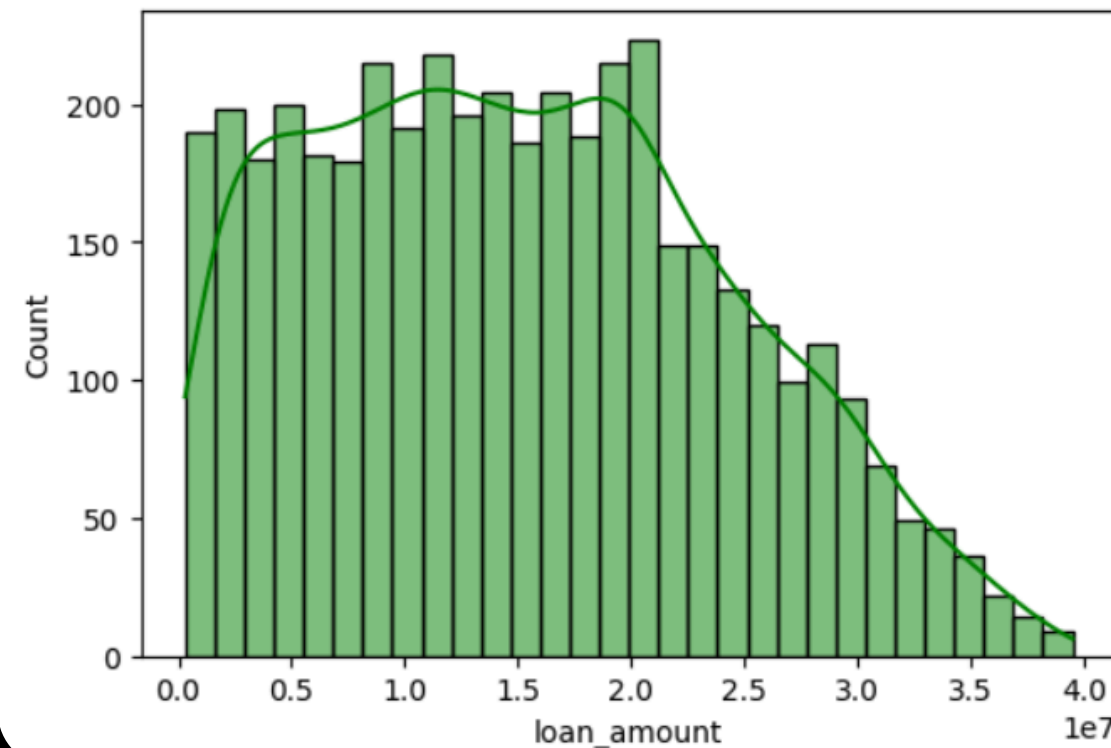
Income vs Loan Status:

The median annual income for both approved and rejected applicants is approximately ₹50 lakh, suggesting income alone has limited predictive power.

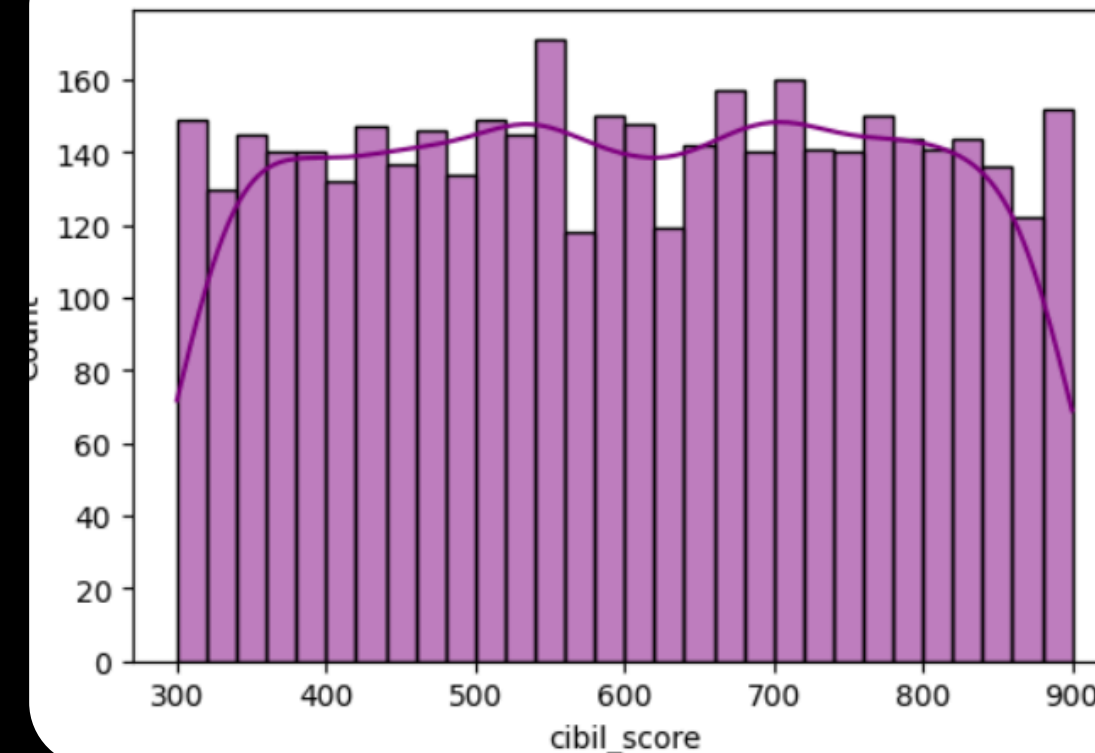
Loan Status Distribution:

Out of roughly 4,300 applications, about 2,650 (62%) were approved while 1,650 (38%) were rejected.

Loan Amount Distribution



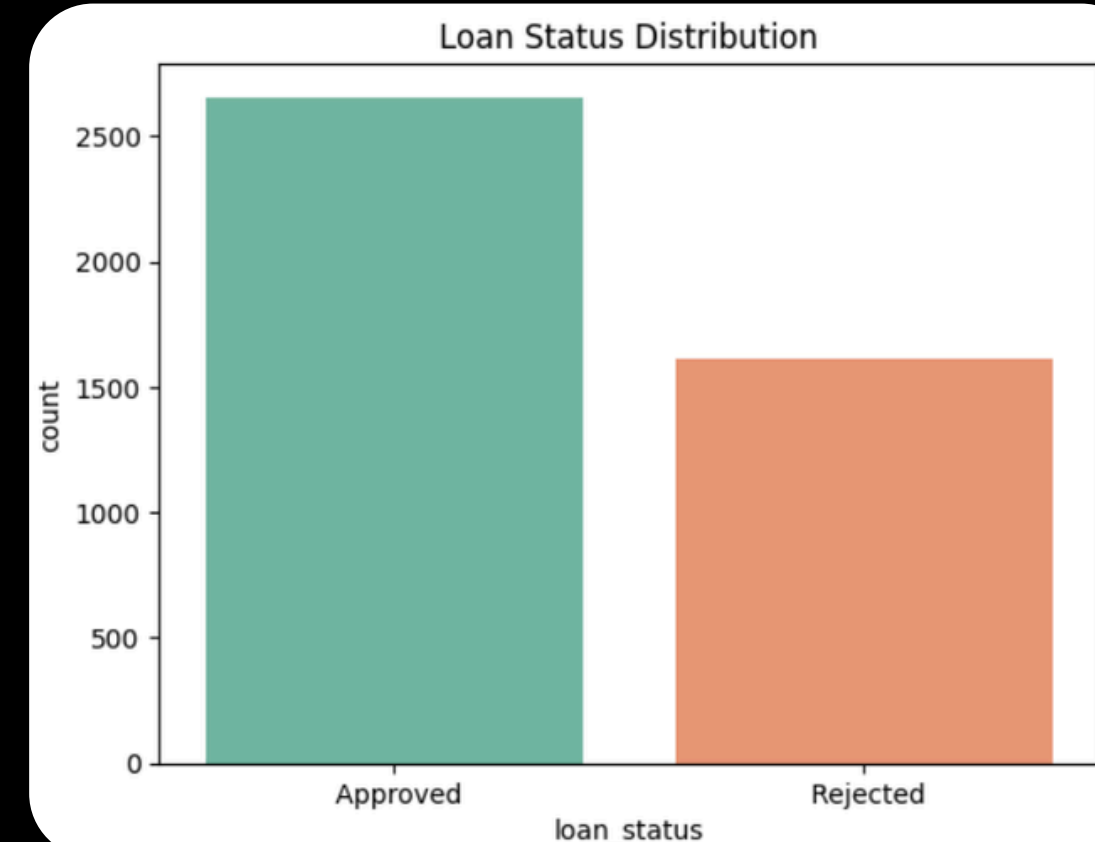
CIBIL Score Distribution



Income vs Loan Status



Loan Status Distribution



Feature Engineering

Engineered Features Overview

Creating required columns

```
df['Debt to income ratio'] = df['loan_amount'] / df['income_annum']
```

```
df['Debt to income ratio']
```

```
: df['total_assets'] = ( df['residential_assets_value'] +  
                        df['commercial_assets_value'] +  
                        df['luxury_assets_value'] +  
                        df['bank_asset_value'])  
  
df['total_assets']
```

Encode categorical variables

```
from sklearn.preprocessing import LabelEncoder  
  
le = LabelEncoder()  
for col in ['education', 'self_employed', 'loan_status']:  
    if col in df.columns:  
        df[col] = le.fit_transform(df[col])  
    else:  
        print(f"Column '{col}' not found")
```

Debt-to-Income

Total Assets

Financial Profile

Model Power

Logistic Regression Model Building

Model Selection & Training Setup

settings features & targets

```
feature_cols = ['no_of_dependents', 'education', 'self_employed', 'income_annum',  
                'loan_amount', 'loan_term', 'cibil_score', 'residential_assets_value', 'commercial_assets_value',  
                'luxury_assets_value', 'bank_asset_value', 'Debt to income ratio', 'total_assets']  
  
X = df[feature_cols]  
y = df['loan_status']
```

Train Test Split

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=18, stratify=y)
```

Scaling features

```
scaler = StandardScaler()  
X_train = scaler.fit_transform(X_train)  
X_test = scaler.transform(X_test)
```

Logistic regression

```
[22]: log_reg = LogisticRegression(max_iter=1000, class_weight="balanced")  
      log_reg.fit(X_train, y_train)
```

```
[22]: ▼ LogisticRegression ⓘ ?  
      ► Parameters
```

Prediction & probability

```
[23]: y_pred = log_reg.predict(X_test)  
      y_proba = log_reg.predict_proba(X_test)[: ,1]
```

80:20 Split

Class Weights

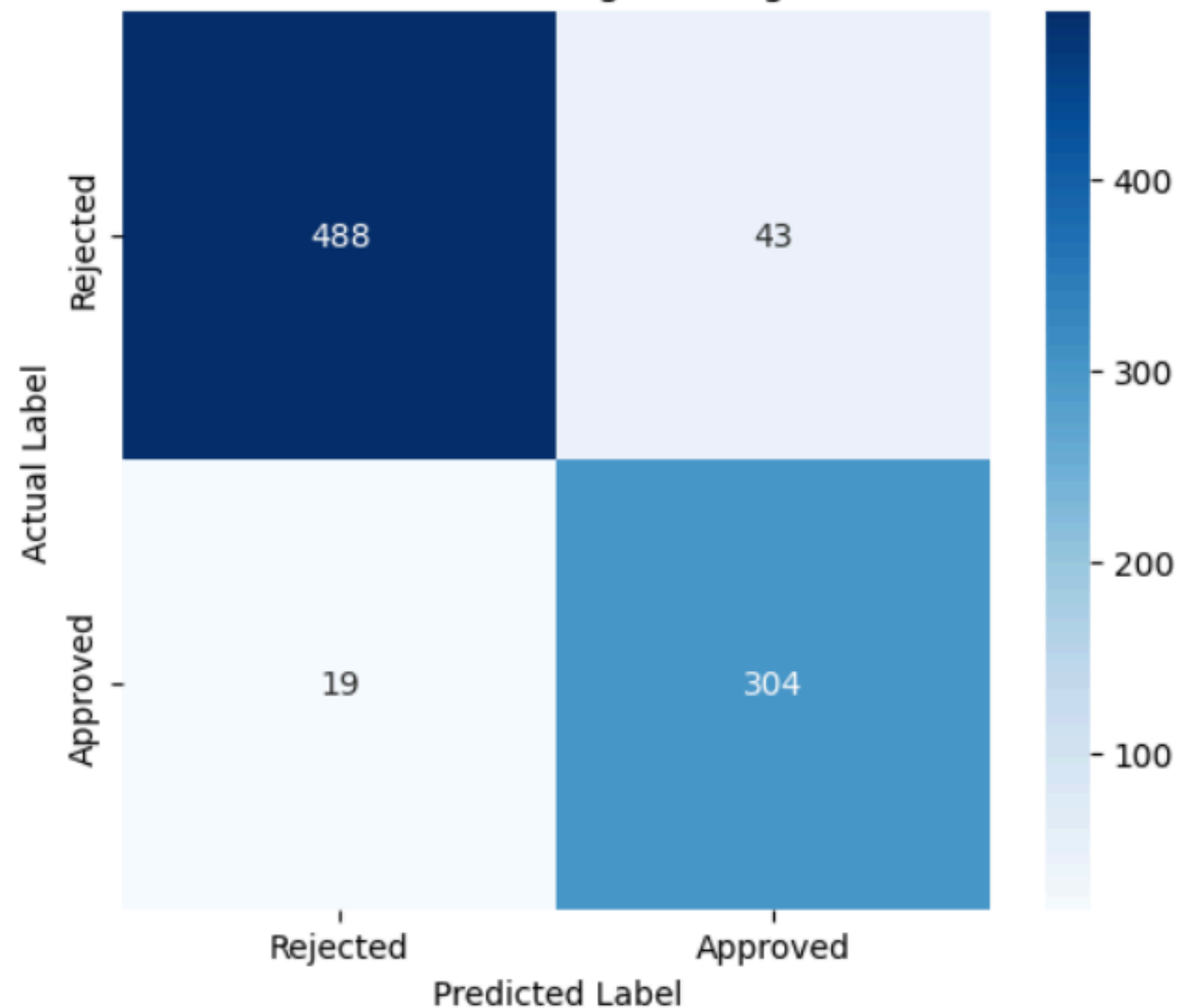
StandardScaler

Binary Output

Model Evaluation

Key Performance Metrics

Confusion Matrix - Logistic Regression



Accuracy: 0.927400468384075
Precision: 0.8760806916426513
Recall: 0.9411764705882353
F1 Score: 0.9074626865671642
ROC-AUC: 0.9659559333694822

Confusion Matrix:

```
[[488  43]
 [ 19 304]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.96	0.92	0.94	531
1	0.88	0.94	0.91	323
accuracy			0.93	854
macro avg	0.92	0.93	0.92	854
weighted avg	0.93	0.93	0.93	854

Accuracy: 92.7%

ROC-AUC: 96.6%

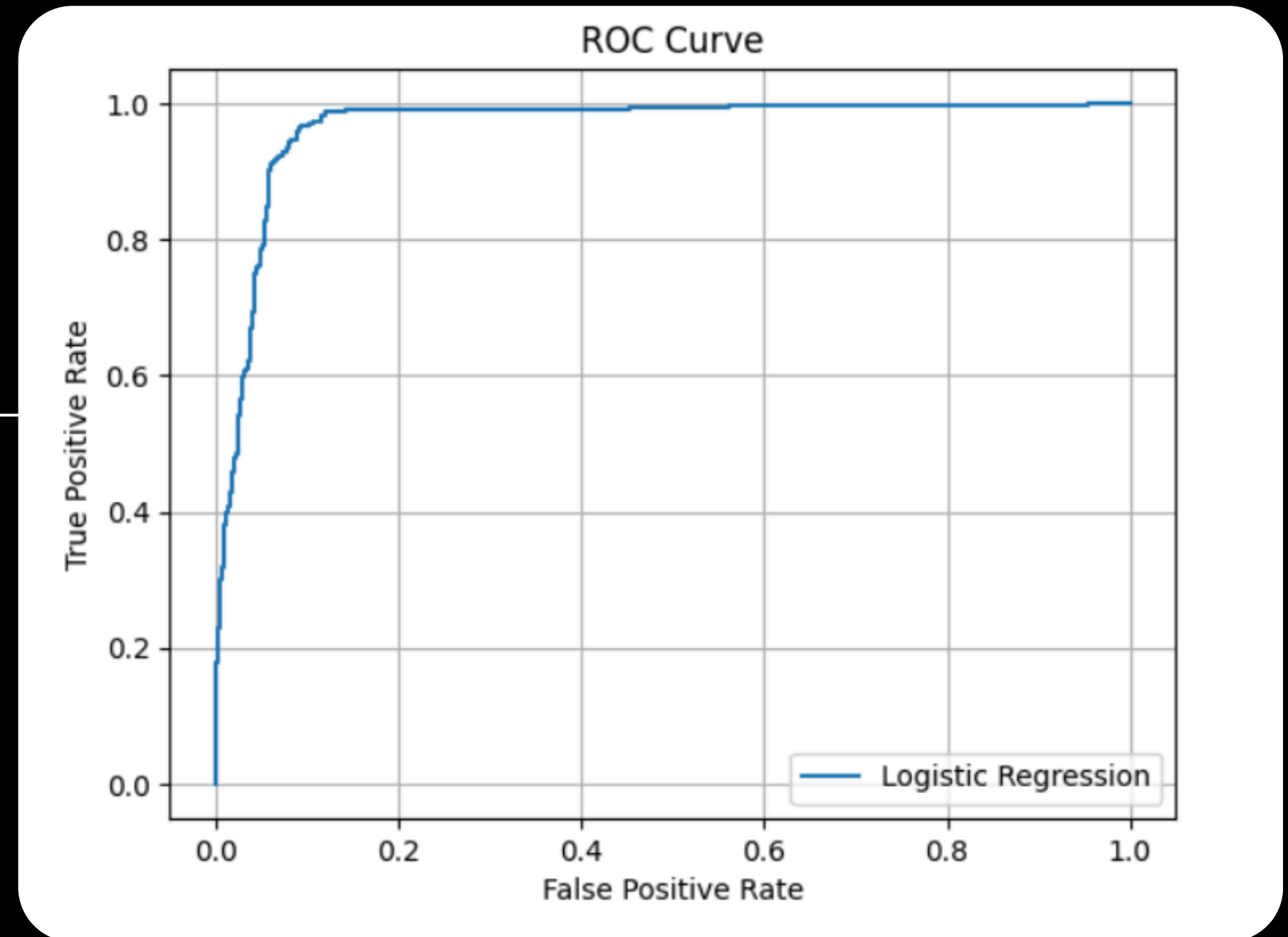
Precision: High

F1 Score: Strong

ROC Curve Performance Analysis

ROC-AUC Score: 96.6%

The ROC curve measures classification quality across all decision thresholds. A higher curve indicates superior model discrimination between approved and rejected loans, demonstrating exceptional predictive performance.



Key Findings & Business Impact

Top Predictive Factors

Key Findings

1. Achieved 92.7% prediction accuracy using Logistic Regression for loan approval classification.
2. Credit score, applicant income, and debt-to-income ratio were the strongest predictors of approval.
3. The model demonstrated excellent discrimination ability with a 96.6% ROC-AUC score.
4. Feature engineering improved model performance and enhanced the understanding of applicant risk profiles.

Business Impact

- Enables faster loan eligibility assessment through automated decision support.
- Helps financial institutions reduce credit risk by identifying potentially risky applicants.
- Improves consistency by supporting data-driven loan approval decisions.
- Increases operational efficiency by reducing manual effort in the loan screening process.

Reduced Risk

Consistency

Faster Approvals

Better Experience

Conclusion

Final Conclusion (Key Points)

- Developed a Loan Approval Prediction Model that achieved 92.7% accuracy, demonstrating strong predictive performance.
- Identified credit score, income, and debt-to-income ratio as the most influential factors in loan approval decisions.
- Enabled data-driven risk assessment, helping lenders distinguish between eligible and high-risk applicants.
- Demonstrated the potential of machine learning to streamline loan approvals, reduce credit risk, and improve operational efficiency in financial institutions.

92.7% Accuracy

Suggestions

- Incorporate additional applicant attributes such as employment history, loan purpose, and repayment behavior to improve prediction accuracy.
- Experiment with advanced machine learning models like Random Forest, XGBoost, or Gradient Boosting for potential performance gains.
- Deploy the model as a real-time loan eligibility screening tool to support faster decision-making.
- Continuously retrain and monitor the model with new loan application data to maintain accuracy and adapt to changing lending patterns.

96.6% ROC-AUC

Thank You

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Portfolio website

